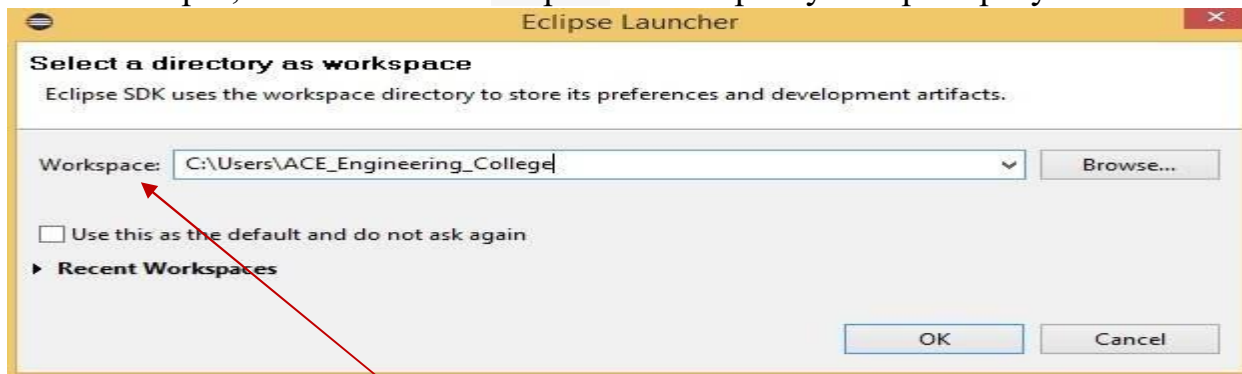


Experiment No. (1):

Use Eclipse or net bean platform and acquaint with the various menus. Create a test project, add test class, and run it. See how you can use auto suggestions, auto fill. Try code formatter and code refactoring like renaming variables, methods and classes. Try debug step by step with a small program of about 10 to 15 lines which contains at least one if else condition and a for loop.

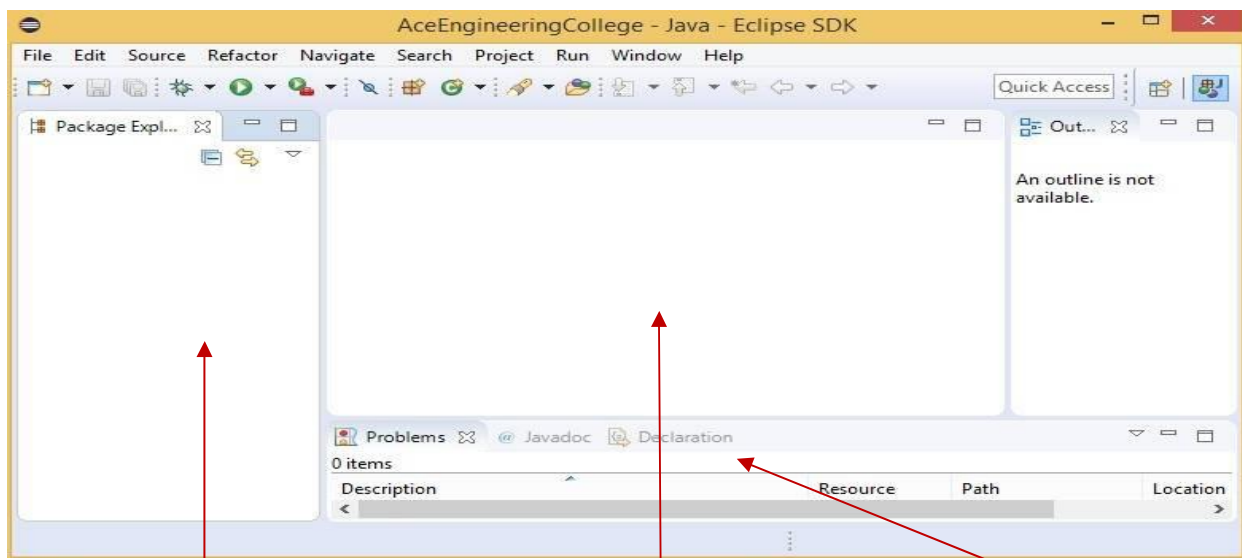
Starting the Eclipse IDE

To start Eclipse, double-click the eclipse. The Eclipse system prompts you for a *workspace*.



The workspace is the physical location you are storing certain Meta data, source files, images etc

Eclipse starts and shows the *Welcome* page. After closing the welcome screen, the Eclipse Layout should look similar to the following screenshot.



Package Explorer views allows you to browse the structure of your project.

JAVA Editor view is used to modify java source code. Each java source file is opened in a

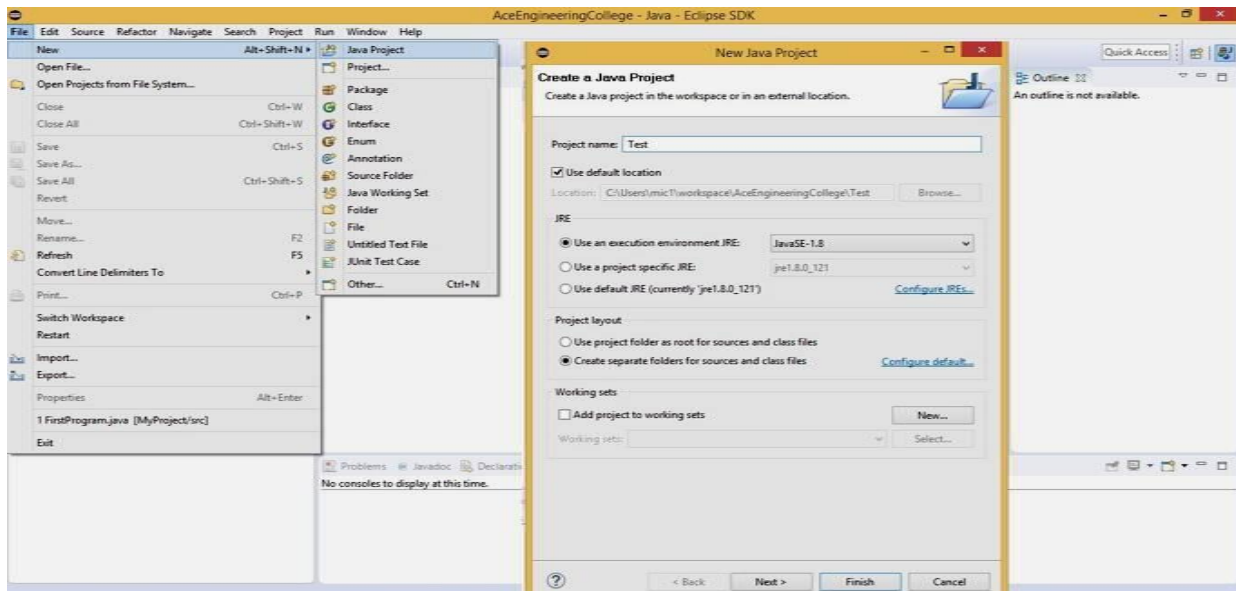
Output console displays the output of the program and Problem view shows the errors and warning

Create your first Java program

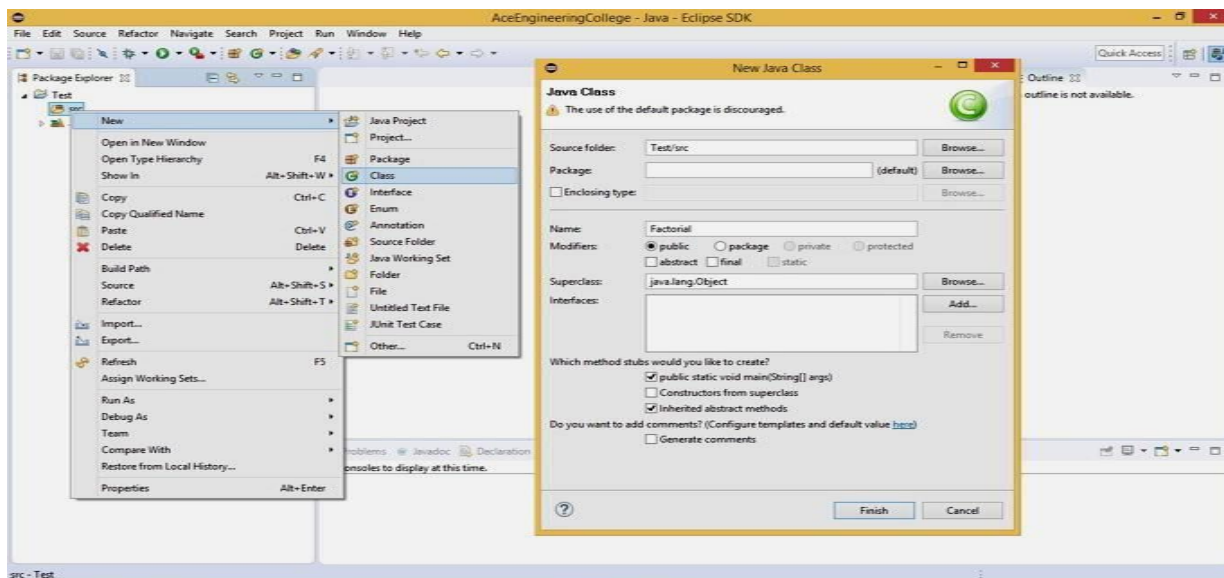
The following section describes how to create a Java application using Eclipse. We will write a program that will print “Factorial of a number” to the console.

Create project

Select **File ► New ► Java project** from the menu. Enter *Test* as the project name and press **Finish** the button to create the project.

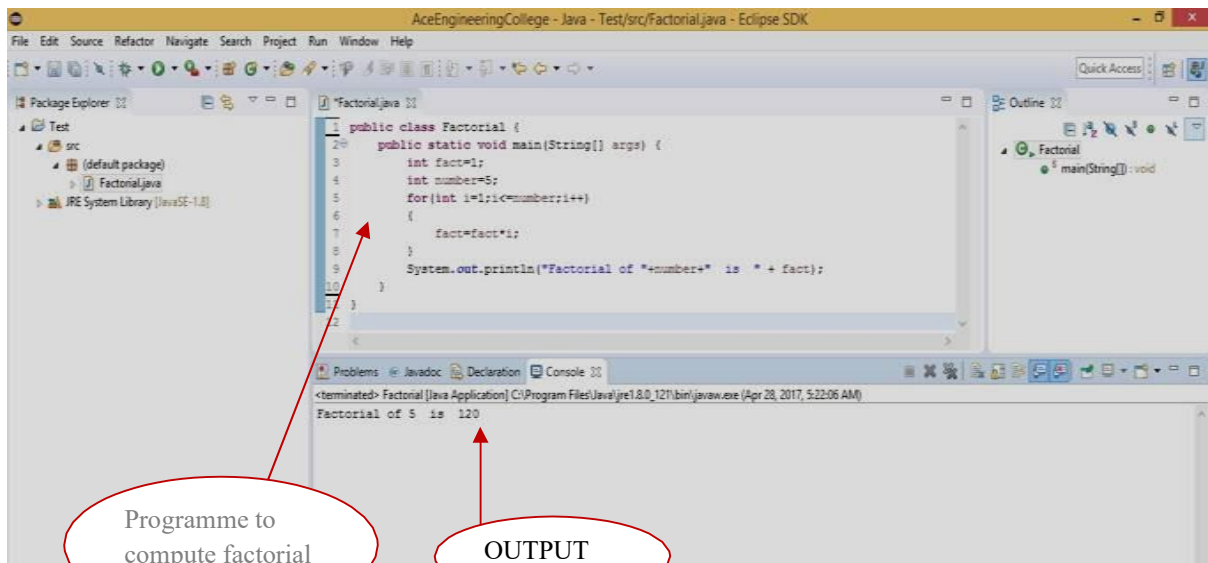


Create Java class



Right-click on your Project and select **New ► Class** to create a Java class.

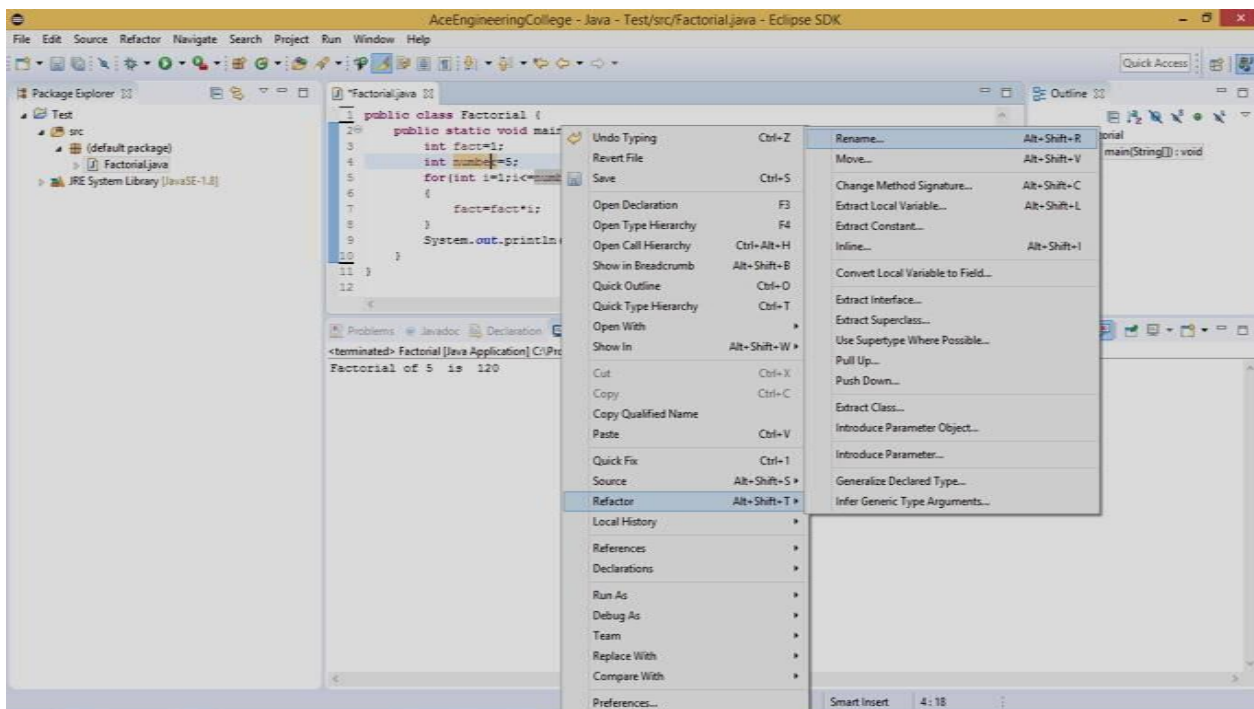
Enter *Factorial* as the class name and select the ***public static void main (String[] args)*** checkbox. Press the **Finish** button. This creates a new class file and opens the Java Editor.



Code refactoring like renaming variables, methods and classes:

To rename a variable *number* in the above program, just right click on the variable name go to *Refactor*

and click Rename. Name of variable *number* will change in the entire project.



IfElseIfExample.java

```
public class IfElseIfExample
{
    public static void main(String[] args)
    {
        int marks=65;
        if(marks<50)
        {
            System.out.println("fail");
        }
        else if(marks>=50 && marks<60)
        {
            System.out.println("D grade");
        }
        else if(marks>=60 && marks<70)
        {
            System.out.println("C grade");
        }
        else if(marks>=70 && marks<80)
        {
            System.out.println("B grade");
        }
        else if(marks>=80 && marks<90)
        {
            System.out.println("A grade");        }
        else if(marks>=90 && marks<100)
        {
            System.out.println("A+ grade");        }
        else
        {
            System.out.println("Invalid!");        }
    }
}
```

Viva Questions:

1. What do you mean by Platform Independence?

Platform Independence you can run and compile program in one platform and can execute in any other platform.

2. What is JIT Compiler?

Just-In-Time(JIT) compiler is used to improve the performance. JIT compiles parts of the bytecode that have similar functionality at the same time

3. What all memory areas are allocated by JVM?

Heap, Stack, Program Counter Register and Native Method Stack

4. What is the base

class of all classes?

java.lang.Object

5. What are two different ways to call

garbage collector? System.gc() OR

Runtime.getRuntime().gc().

6. Use of finalize() method in java?

finalize() method is used to free the allocated resource.

7. List two java IDEs?

1.Eclipse, 2.Net beans and 3.IntelliJ

8. What are java buzzwords?

Java buzzwords explain the important features of java. They are Simple, Secured, Portable, architecture neutral, high performance, dynamic, robust, interpreted etc.

9. Is byte code similar to .obj file in C?

Yes, both are machine understandable codes No, .obj file directly understood by machine, byte code requires JVM.

10. What are length and length() in Java?

Both give number of char/elements, length is variable defined in Array class, length() is method defined in String class.